

MALIK JIRASEK

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Institute for Theoretical Physics
University of Tübingen, Germany
Auf der Morgenstelle 14
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EDUCATION

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| Oct 2025 – Present | Master of Science in Physics, University of Tübingen, Germany
Thesis: "Keldysh-field-theory approach to the quantum Fisher information"
Supervisor: Igor Lesanovsky |
| Aug 2020 – Jul 2025 | Bachelor of Science in Physics, University of Tübingen, Germany
Final grade: 1.04 / 5.0 (German system; 1.0 = excellent)
Thesis: "Quantum Sensing beyond the Heisenberg Limit with the Boundary Time Crystal"
Supervisor: Igor Lesanovsky |

RESEARCH EXPERIENCE

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| Nov 2025 – Present | Master of Science Thesis Research, University of Tübingen, Germany <ul style="list-style-type: none">Developing a Keldysh-field-theory description of the quantum Fisher information in open quantum systems, with a focus on scalable sensor architectures.Results incorporated into a manuscript submitted to <i>Physical Review Letters</i> (under review). |
| Aug 2025 – Sep 2026 | Working Student, Bosch eBike Systems, Germany <ul style="list-style-type: none">Implemented a Machine Learning pipeline for battery cell aging predictions, using BiLSTM networks and TensorFlow.Collaborated with the engineering team to integrate the model into the production workflow. |
| Oct 2024 – Jul 2025 | Bachelor of Science Thesis Research, University of Tübingen, Germany <ul style="list-style-type: none">Demonstrated collective enhancement of sensitivity bounds for measuring optical phases using a boundary time crystal as light source.Results incorporated into a paper published in <i>Physical Review A</i>. |
| Aug 2024 – Oct 2024 | Research Intern, Humboldt University of Berlin, Germany <ul style="list-style-type: none">Extended a Fortran-based many-body solver to include high-precision interaction potentials for ultracold-atom tweezer arrays, achieving 10^{-8} relative precision in ground state energies. |

PUBLICATIONS

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| Preprint | Malik Jirasek , Igor Lesanovsky, and Viktoria Noel, "Quantum Fisher information of driven-dissipative systems from Keldysh path integrals". arXiv:2609.26462 [quant-ph]. |
| Published | Malik Jirasek , Igor Lesanovsky, and Albert Cabot, "Boundary time crystal as a light source for collectively enhanced sensing", <i>Phys. Rev. A</i> 114 , 032611(2026). |

PRESENTATIONS & CONFERENCES

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| 20 Oct 2026 | Invited for a talk at the Cold Quantum Coffee, Heidelberg |
| 30 Sep 2026 | Presented poster at DQUT Conference, Palma de Mallorca |
| 07 Sep 2026 | Presented poster at QUANT26 Master School, Dresden |
| 06 Mar 2026 | Contributed Talk at DPG Spring Meeting SAMOP, Mainz |

TEACHING EXPERIENCE

WS 2024/25	Advanced Quantum Mechanics tutor (University of Tübingen)
SS 2024	Quantum Mechanics tutor (University of Tübingen)
WS 2023/24	Analytical Mechanics tutor (University of Tübingen)
SS 2023	Physics 2 tutor (University of Tübingen)
WS 2022/23	Physics 1 tutor (University of Tübingen)
WS 2021/22 - WS 2024/25	Physics Lab tutor (University of Tübingen)

AWARDS & SCHOLARSHIPS

2021 & 2023	Deutschlandstipendium (merit-based; awarded to ~0.5% of enrolled students)
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SKILLS & METHODS

Programming	Scientific programming, high performance computing, machine learning methods
Simulation & Modeling	Stochastic calculus, nonequilibrium statistical mechanics, many-body systems
Soft Skills	Communication between different scientific disciplines, teamwork, problem-solving
Languages	German (native), English (full professional proficiency), French (beginner)

SERVICE & LEADERSHIP

Apr 2027	Co-organizer, Workshop "Dynamics in Strongly Correlated Quantum Open Systems", Tübingen
2026	Student representative, faculty appointment committee (ML in physics)
2022 - 2025	Treasurer & member, physics student council Tübingen